Simulation of affine forward variance models

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April 14, 2021

Abstract

We propose an Euler-like and a hybrid scheme to simulate affine forward variance models.

1 The model

In [GKR19], it is shown that affine forward variance (AFV) models must take the form

$$d\xi_t(u) = g(u-t)\ \sqrt{v_t}\, dW_t$$

so

$$\begin{aligned} v_T = \xi_T(T) &= \xi_0(T) + \int_0^T d\xi_s(T) \\ &= \xi_0(T) + \int_0^T g(T-s)\ \sqrt{v_s}\, dW_s. \end{aligned} \tag{1.1}$$

As in [BLP17], we assume that for some $H \in (0,1)$,

$$g(\tau) = \tau^{H-1/2}\, L_g(\tau), \tau \in (0,T]$$

where $L_g : (0,T] \to [0,\infty)$ is continuously differentiable, slowly varying at 0 and bounded away from 0. Proposition 2.2 of [BLP17] shows that the variogram of v behaves as Δ^H for small Δ, in other words that v has *roughness* H.

1.1 Moments of AFV models

By definition $\mathbb{E}_t\,[v_T] = \xi_t(T)$. Also Itô's isometry gives

$$\mathbb{E}_t\left[(v_T - \xi_t(T))^2\right] = \int_t^T \xi_t(s)\, g(T-s)^2\, ds. \tag{1.2}$$

Thus, we may in principle simulate one step of the variance process $v_t \mapsto v_{t+\Delta}$.

To compute higher central moments of v_T, we may use Theorem 1.2 of [FGR20] according to which the cumulant generating function of v_T is given by the forest expansion

$$\log \mathbb{E}\,[e^{a\,v_T}|\,\mathcal{F}_t] = \sum_{n=1}^{N} z^n \mathbb{K}_t^n(T) + o(|z|^N) \quad \text{as } z \to 0\ ,$$

where the $\mathbb{K}^n$ satisfy the recursion

$$\mathbb{K}_t^1(T) := \mathbb{E}\,[A_T|\,\mathcal{F}_t] \quad \text{and} \quad \forall n > 0: \ \ \mathbb{K}_t^{n+1}(T) = \frac{1}{2}\sum_{k=1}^{n} (\mathbb{K}^k \diamond \mathbb{K}^{n+1-k})_t(T). \tag{1.3}$$

Thus the t-conditional cumulants of v_T are given by $\kappa_n = n!\,\mathbb{K}_t^{(n)}$. With $\mathbb{K}^1 = Y = \xi(T) \equiv \circ$, the first few forests are given by

$$\begin{aligned}
\mathbb{K}^2 &= \tfrac{1}{2} Y \diamond Y \equiv \tfrac{1}{2}\\
\mathbb{K}^3 &= \tfrac{1}{2}(Y \diamond Y) \diamond Y \equiv \tfrac{1}{2}\\
\mathbb{K}^4 &= \tfrac{1}{8}(Y \diamond Y)^{\diamond 2} + \tfrac{1}{2}((Y \diamond Y) \diamond Y) \diamond Y \equiv \tfrac{1}{8} + \tfrac{1}{2}
\end{aligned} \tag{1.4}$$

$$\mathbb{K}^5 = \quad \dots \quad = \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{2} \tag{1.5}$$

To ease notation, define $\mu_n = \mathbb{E}_t\left[(v_T - \xi_t(T))^n\right]$. Then from the well-known relationships between central moments and cumulants, the next few central moments are given by

$$\begin{aligned}
\mu_3 &= \kappa_3 = 3\\
\mu_4 &= \kappa_4 + 3\,\kappa_2^2 = 3 + 12 + 3\,()^2\\
\mu_5 &= \kappa_5 + 10\,\kappa_3\,\kappa_2 = 30 + 15 + 60 + 30\,\left(\right)\,()\,.
\end{aligned}$$

Similarly, $\mu_6 = \kappa_6 + 15\,\kappa_4\,\kappa_2 + 10\,\kappa_3^2 + 15\,\kappa_2^3$.

1.1.1 Tree computations

Assuming the variance process (1.1), we have

$$= \int_t^T \xi_t(s_1)\, g(T-s_1)^2\, ds_1$$

$$= \int_t^T \xi_t(s_2)\, g(T-s_2)\, ds_2 \int_{s_2}^T g(s_1-s_2)\, g(T-s_1)^2\, ds_1 =: \int_t^T \xi_t(s_2)\, g(T-s_2)\, \tilde{g}_{12}(T-s_2)\, ds_2$$

$$= \int_t^T \xi_t(s_2)\, ds_2 \left(\int_{s_2}^T g(s_1-s_2)\, g(T-s_1)^2\, ds_1 \right)^2 = \int_t^T \xi_t(s_2)\, \tilde{g}_{12}(T-s_2)^2\, ds_2$$

$$= \int_t^T \xi_t(s_3)\, g(T-s_3)\, ds_3 \int_{s_3}^T g(s_2-s_3)\, g(T-s_2)\, ds_2 \int_{s_2}^T g(s_1-s_2)\, g(T-s_1)^2\, ds_1$$

$$=: \int_t^T \xi_t(s_3)\, g(T-s_3)\, \tilde{g}_{11,12}(T-s_3)\, ds_3$$

$$= \int_t^T \xi_t(s_3)\, ds_3 \int_{s_3}^T g(s_1-s_3)\, g(T-s_1)^2\, ds_1 \int_{s_3}^T g(s_2-s_3)\, g(T-s_2)\, \tilde{g}_{12}(T-s_2)\, ds_2$$

$$= \int_t^T \xi_t(s_3)\, \tilde{g}_{12}(T-s_3)\, \tilde{g}_{11,12}(T-s_3)\, ds_3$$

$$= \int_t^T \xi_t(s_3)\, g(T-s_3)\, ds_3 \int_{s_3}^T g(s_2-s_3)\, \tilde{g}_{12}(T-s_2)^2\, ds_2$$

$$= \int_t^T \xi_t(s_4)\, g(T-s_4)\, ds_4 \int_{s_4}^T g(s_3-s_4)\, g(T-s_3)\, \tilde{g}_{11,12}(T-s_3)\, ds_3.$$

1.1.2 Power-law kernel

We now compute trees in the special case with

$$g(\tau) = \sqrt{2H}\, \eta\, \tau^{\alpha-1} =: \tilde{\eta}\tau^{\alpha-1}, \tag{1.6}$$

where $\alpha = H + \frac{1}{2}$. Computation in this special case is substantially simplified by noting that trees have a simple form, as given in the following lemma.

Lemma 1.1. *In the AFV model with the power-law kernel* (1.6)*, diamond trees with k leaves, and hence all forests terms $\mathbb{K}_t^k$ in* (1.3) *are of the form*

$$C_k \int_t^T \xi_t(s)\, (T-s)^{2H(k-1)-1}\, ds. \tag{1.7}$$

Moreover, root-joining trees with i and j leaves results in a tree with $k = i + j$ leaves and a prefactor given by

$$C_1 = 1, \quad C_k = \tilde{\eta}^2\, C_i\, C_j\, \beta(H+\tfrac{1}{2}, 2H(i-1))\, \beta(H+\tfrac{1}{2}, 2H(j-1)),\ k > 1.$$

Proof. A simple computation gives

$$\text{} = \tilde{\eta}^2 \int_t^T \xi_t(s_1)\,(T-s_1)^{2H-1}\,ds_1$$

which is consistent with the claimed form for diamond trees with two leaves. We now proceed by induction. Consider two trees with i and j leaves respectively. Then according to the induction hypothesis,

$$\mathbb{T}^i_s = C_i \int_s^T \xi_s(s_1)\,(T-s_1)^{2H(i-1)-1}\,ds_1; \quad \mathbb{T}^j_s = C_j \int_s^T \xi_s(s_2)\,(T-s_2)^{2H(j-1)-1}\,ds_2.$$

Then, recalling that $\alpha = H + \frac{1}{2}$,

$$\begin{aligned}
\left(\mathbb{T}^i \diamond \mathbb{T}^j\right)_t(T) &= \mathbb{E}_t\left[\int_t^T d\langle \mathbb{T}^i, \mathbb{T}^j\rangle_s\right] \\
&= \tilde{\eta}^2\, C_i\, C_j \int_t^T \xi_t(s)\,ds \int_s^T (s-s_1)^{\alpha-1}\,(T-s_1)^{2H(i-1)-1}\,ds_1 \\
&\quad \times \int_s^T (s-s_2)^{\alpha-1}\,(T-s_2)^{2H(j-1)-1}\,ds_2 \\
&= \tilde{\eta}^2\,\beta(H+\tfrac{1}{2}, 2H(i-1))\,\beta(H+\tfrac{1}{2}, 2H(j-1))\,C_i\,C_j \int_t^T \xi_t(s)\,(T-s)^{2H(i+j-1)-1}\,ds,
\end{aligned}$$

and the induction step is complete. □

Remark 1.1. *Note that trees with k leaves are of order* $\left((T-t)^{2H}\right)^{k-1}$, $k > 1$.

Let

$$J^k_t(T) = \int_t^T \xi_t(s)\,(T-s)^{2H(k-1)-1}\,ds,$$

and let $B_j = \beta(H+\frac{1}{2}, 2H(j-1))$. Note in particular that $B_1 = 1$. Then under root-joining, with $k = i + j$, $C_k = \tilde{\eta}^2\, C_i\, C_k\, B_i\, B_j$. Applying this rule gives

$$\begin{aligned}
\text{ } &= \tilde{\eta}^2\, J^2_t(T) \\
\text{ } &= \tilde{\eta}^4\, B_2\, J^3_t(T) \\
\text{ } &= \tilde{\eta}^6\, B_2^2\, J^4_t(T) \\
\text{ } &= \tilde{\eta}^6\, B_2\, B_3\, J^4_t(T) \\
\text{ } &= \tilde{\eta}^8\, B_2^2\, B_3\, J^5_t(T) \\
\text{ } &= \tilde{\eta}^8\, B_2^2\, B_4\, J^5_t(T) \\
\text{ } &= \tilde{\eta}^8\, B_2\, B_3\, B_4\, J^5_t(T).
\end{aligned}$$

1.1.3 Scaling of central moments

According to our assumption,

$$g(\tau) = \tilde{\eta}\,\tau^{H-1/2}\,L_g(\tau), \tau \in (0, T]$$

where $L_g : (0, T] \to [0, \infty)$ is a slowly varying function. The generic form of trees in Section 1.1.1 and Lemma 1.1 then imply that a generic tree with k leaves may be approximated in the limit $T \to t$ as

$$\begin{aligned} L_g(T)^{2k-1}\,\mathbb{T}_t^k(T) &= L_g(T)^{2k-1}\,\xi_t(T)\,C_k \int_t^T (T-s)^{2H(k-1)-1}\,ds \\ &= L_g(T)^{2k-1}\,\xi_t(T)\,C_k\,\frac{1}{2H(k-1)}\,(T-t)^{2H(k-1)}. \end{aligned}$$

From (1.3), the cumulants κ_n are given by forests of trees with n leaves. Thus κ_n scales as $(T-t)^{2H(n-1)}$ as $T \to t$. Finally, central moments are related to cumulants by the formula[1]

$$\mu_n = \sum_{k=1}^{n} B_{n,k}(0, \kappa_2, ..., \kappa_{n-k+1})$$

where the $B_{n,k}(\cdot)$ are Bell polynomials. One property of Bell polynomials is that the subscripts must add up to n, as in $\mu_6 = \kappa_6 + 15\,\kappa_4\,\kappa_2 + 10\,\kappa_3^2 + 15\,\kappa_2^3$.

Abusing notation, put $\tau = T - t$. Then, given that $\kappa_j \sim \tau^{2H(j-1)}$, we see that the individual terms in the expression for μ_n scale as τ^α with $\alpha \le 2H(n-1)$. The minimum value of α depends on whether n is even or odd. If n is even, the Bell polynomial contains the monomial $\kappa_2^{n/2}$ which scales as $\tau^{2Hn/2}$. If n is odd, the Bell polynomial contains a term of the form $\kappa_3\,\kappa_2^{(n-3)/2}$ which scales as $\tau^{4H}\,\tau^{2H(n-3)/2}$. We deduce that in the limit $\tau \to 0$, μ_n scales as a polynomial of terms with powers τ^α with

$$2H\lfloor \tfrac{n+1}{2} \rfloor \le \alpha \le 2H(n-1). \tag{1.8}$$

Applying (1.8) to the first six central moments we see that as $\tau \to 0$,

$$\begin{aligned} \mu_2 &\sim \tau^{2H} \\ \mu_3 &\sim \tau^{4H} \\ \mu_4 &\sim \tau^{4H} \\ \mu_5 &\sim \tau^{6H} \\ \mu_6 &\sim \tau^{6H}. \end{aligned}$$

1.2 Approximation of moments

Define the non-centralized moments

$$\mu_n' = \mathbb{E}_t\left[v_T^n\right].$$

[1] Wikipedia `https://en.wikipedia.org/w/index.php?title=Cumulant&oldid=974339792`.

In an analogous manner, we have

$$\mu'_n = \sum_{k=1}^{n} B_{n,k}(\kappa_1, \kappa_2, ..., \kappa_{n-k+1}).$$

Moreover, since $v_T \geq 0$, $\mu'_n \geq 0$. The first few moments are given by

$$\begin{aligned}
\mu'_1 &= \kappa_1 \\
\mu'_2 &= \kappa_2 + \kappa_1^2 \\
\mu'_3 &= \kappa_3 + 3\kappa_2\kappa_1 + \kappa_1^3 \\
\mu'_4 &= \kappa_4 + 4\kappa_3\kappa_1 + 3\kappa_2^2 + 6\kappa_2\kappa_1^2 + \kappa_1^4 \\
\mu'_5 &= \kappa_5 + 5\kappa_4\kappa_1 + 10\kappa_3\kappa_2 + 10\kappa_3\kappa_1^2 + 15\kappa_2^2\kappa_1 + 10\kappa_2\kappa_1^3 + \kappa_1^5 \\
\mu'_6 &= \kappa_6 + 6\kappa_5\kappa_1 + 15\kappa_4\kappa_2 + 15\kappa_4\kappa_1^2 + 10\kappa_3^2 + 60\kappa_3\kappa_2\kappa_1 + 20\kappa_3\kappa_1^3 \\
&\qquad\qquad + 15\kappa_2^3 + 45\kappa_2^2\kappa_1^2 + 15\kappa_2\kappa_1^4 + \kappa_1^6.
\end{aligned}$$

Once again, for any appropriate kernel function g, the generic form of trees in Section 1.1.1 and Lemma 1.1 imply that a generic tree with k leaves may be approximated in the limit $T \to t$ as

$$L_g(T)^{2k-1}\, \mathbb{T}_t^k(T) \quad = \quad L_g(T)^{2k-1}\, \xi_t(T)\, C_k\, \frac{1}{2H(k-1)}\, (T-t)^{2H(k-1)}.$$

Since all trees in all forests can be computed to order $O(\Delta)$, all the μ'_n may be computed to the same order.

2 Simulation

Without loss of generality, let $t = 0$ and $\xi(u) = \xi_0(u)$. Let $\Delta = T/n$. Following [BLP17], we have the exact decomposition

$$v(n\Delta) = \xi(n\Delta) + \sum_{k=1}^{n} \int_{(k-1)\Delta}^{k\Delta} g(n\Delta - s)\, \sqrt{v_s}\, dW_s$$

For ease of notation, denote $v(n\Delta)$ by v_n and so on. Then

$$v_n = \xi_n + \sum_{k=1}^{n} \int_{(k-1)\Delta}^{k\Delta} g(n\Delta - s)\, \sqrt{v_s}\, dW_s = \hat{\xi}_n + u_n$$

where

$$\hat{\xi}_n = \mathbb{E}\,[\, v_n | \mathcal{F}_{n-1}] = \xi_n + \sum_{k=1}^{n-1} \int_{(k-1)\Delta}^{k\Delta} g(n\Delta - s)\, \sqrt{v_s}\, dW_s$$

and

$$u_n = \int_{(n-1)\Delta}^{n\Delta} g(n\Delta - s)\ \sqrt{v_s}\, dW_s.$$

In order to simulate the stock price process, we also need to simulate[2]

$$\chi_k = \int_{(k-1)\Delta}^{k\Delta} \sqrt{v_s}\, dW_s.$$

2.1 The covariance matrix

At each step of the algorithm, we need to update $v_k, \hat{\xi}_{k+1}, \chi_k$. Let's compute their covariance matrix. Recall that

$$\begin{aligned} v_k &= \hat{\xi}_k + \int_{(k-1)\Delta}^{k\Delta} g(k\Delta - s)\ \sqrt{v_s}\, dW_s \\ \hat{\xi}_{k+1} &= \mathbb{E}\left[v_{k+1} | \mathcal{F}_{k-1}\right] + \int_{(k-1)\Delta}^{k\Delta} g((k+1)\Delta - s)\ \sqrt{v_s}\, dW_s \\ \chi_k &= \int_{(k-1)\Delta}^{k\Delta} \sqrt{v_s}\, dW_s. \end{aligned}$$

Define

$$\mathcal{G}_i(\Delta) = \int_0^\Delta g(s + i\Delta)\, ds; \quad \mathcal{G}_{i,j}(\Delta) = \int_0^\Delta g(s + i\Delta)\, g(s + j\Delta)\, ds \tag{2.1}$$

Then, for suitably defined averages $\bar{\xi}$ of ξ (all different, depending on the case) or alternatively, approximating the forward variance curve as flat over the interval $(0, \Delta]$,

$$\begin{aligned} \mathrm{var}[v_k] &\approx \bar{\xi}\, \mathcal{G}_{0,0}(\Delta) \\ \mathrm{var}[\hat{\xi}_{k+1}] &\approx \bar{\xi}\, \mathcal{G}_{1,1}(\Delta) \\ \mathrm{var}[\chi_k] &\approx \bar{\xi}\, \Delta \\ \mathrm{cov}[v_k, \hat{\xi}_{k+1}] &\approx \bar{\xi}\, \mathcal{G}_{0,1}(\Delta) \\ \mathrm{cov}[v_k, \chi_k] &\approx \bar{\xi}\, \mathcal{G}_0(\Delta) \\ \mathrm{cov}[\chi_k, \hat{\xi}_{k+1}] &\approx \bar{\xi}\, \mathcal{G}_1(\Delta). \end{aligned} \tag{2.2}$$

Given a suitable kernel, all of these may be easily computed.

2.1.1 The correlation matrix

Because variances and covariances in an AFV model are linear in the forward variance curve, and approximating the forward variance curve as flat over the integration interval,

[2]We write the increments as χ_k to emphasize that they should be approximately χ^2 distributed random variables.

the correlation matrix takes a particularly simple form.

$$R = \begin{pmatrix} 1 & \rho_{v\xi} & \rho_{v\chi} \\ \rho_{v\xi} & 1 & \rho_{\xi\chi} \\ \rho_{v\chi} & \rho_{\xi\chi} & 1 \end{pmatrix} \tag{2.3}$$

where all of these entries are *independent* of the time step k. Specifically,

$$\rho_{v\xi} = \frac{\mathcal{G}_{0,1}(\Delta)}{\sqrt{\mathcal{G}_{0,0}(\Delta)}\sqrt{\mathcal{G}_{1,1}(\Delta)}}; \quad \rho_{v\chi} = \frac{\mathcal{G}_0(\Delta)}{\sqrt{\Delta}\sqrt{\mathcal{G}_{0,0}(\Delta)}}; \quad \rho_{\xi\chi} = \frac{\mathcal{G}_1(\Delta)}{\sqrt{\Delta}\sqrt{\mathcal{G}_{1,1}(\Delta)}}.$$

2.1.2 A further approximation for small H

We now note that in the case of a kernel with a singularity at 0, we have

$$\mathcal{G}_{0,1}(\Delta) \approx \frac{1}{\Delta}\mathcal{G}_1(\Delta)\mathcal{G}_0(\Delta); \quad \mathcal{G}_{1,1}(\Delta) \approx \frac{1}{\Delta}\mathcal{G}_1(\Delta)^2. \tag{2.4}$$

Plugging these approximations into (2.2) gives the correlation matrix

$$R = \begin{pmatrix} 1 & \rho & \rho \\ \rho & 1 & 1 \\ \rho & 1 & 1 \end{pmatrix} \tag{2.5}$$

where[3]

$$\rho \approx \frac{\mathcal{G}_0(\Delta)}{\sqrt{\mathcal{G}_{0,0}(\Delta)\,\Delta}}.$$

Note that in this approximation, given $\hat{\xi}_{k+1}, \chi_1$ is fixed.

Example 2.1. *Power-law kernel*

With the power-law kernel, the following ratios are independent of Δ*. With* $\alpha = 0.55$*,*

$$\frac{\mathcal{G}_{0,1}(\Delta)}{\frac{1}{\Delta}\mathcal{G}_1(\Delta)\mathcal{G}_0(\Delta)} = 1.0496; \quad \frac{\mathcal{G}_{1,1}(\Delta)}{\frac{1}{\Delta}\mathcal{G}_1(\Delta)^2} = 1.00808.$$

and with $\alpha = 0.51$*,*

$$\frac{\mathcal{G}_{0,1}(\Delta)}{\frac{1}{\Delta}\mathcal{G}_1(\Delta)\mathcal{G}_0(\Delta)} = 1.06085; \quad \frac{\mathcal{G}_{1,1}(\Delta)}{\frac{1}{\Delta}\mathcal{G}_1(\Delta)^2} = 1.00959.$$

Thus, at least numerically, the approximations appear good.

3 Positive simulation schemes

Obviously, depending on how we sample the random variables u_n, we could have $\hat{v}_n < 0$, which is obviously undesirable and inconsistent with the process being simulated. We therefore focus on two schemes that guarantee positivity of $\hat{v}_n$.

[3]This is exactly the same expression for ρ that appears in [BLP17].

3.1 The Andersen Quadratic Exponential (QE) scheme

Let us $\hat{\xi}_n = \mathbb{E}[v_n|\mathcal{F}_{n-1}]$. FollowingAndersen's QE scheme as explained in [Alf15], we simulate differently according to the value of the ratio

$$\psi_n := \frac{\mathrm{var}[v_n|\mathcal{F}_{n-1}]}{\hat{\xi}_n^2}.$$

Algorithm ψ^-

If $\psi_n \leq 2$, simulate v_n as

$$v_n = \alpha_n \, (\beta_n + Z_n)^2$$

with $Z_n \sim N(0,1)$ and

$$\beta_n^2 = \frac{2}{\psi_n} - 1 + \sqrt{\frac{2}{\psi_n}} \sqrt{\frac{2}{\psi_n} - 1}; \quad \alpha_n = \frac{\hat{\xi}_n}{1+\beta_n^2}.$$

Algorithm ψ^-

On the other hand, if $\psi_n \geq 1$, simulate v_n as

$$v_n = -\mathbb{1}_{\{U_n < p_n\}} \, \gamma_n \, \log \frac{U_n}{p_n}$$

with $U_n \sim \mathcal{U}(0,1)$ and

$$p_n = \frac{2}{1+\psi_n}; \quad \gamma_n = \frac{1}{2} \hat{\xi}_n \, (1+\psi_n).$$

It is straightforward to check that means and variances are correctly matched in both cases. Since the two regions of applicability overlap, Andersen suggests to use algorithm ψ^- if $\psi_n < 3/2$ and algorithm ψ^+ if $\psi_n \geq 3/2$.

Remark 3.1. *In the classical Heston case, there is would be nothing to prevent us using the more accurate Ninomya-Victoir step in the case $\psi_n \leq 2$.*

3.2 The discrete scheme of Mackevičius

Theorem 3 of [Mac11] gives the potentially first order weak scheme

$$v_n = v^{\pm} = \hat{\xi}_n \left[1 + \psi_n \pm \sqrt{(1+\psi_n)\,\psi_n} \right],$$

where as before

$$\psi_n = \frac{\mathrm{var}[v|\mathcal{F}_{n-1}]}{\hat{\xi}_n^2}.$$

and the probabilities $p^{\pm} := \mathbb{P}[v_n = v^{\pm}]$ are given by

$$p^{\pm} = \frac{1}{2\sqrt{(1+\psi_n)}} \frac{1}{\sqrt{1+\psi_n} \pm \sqrt{\psi_n}}.$$

Note that if $\hat{\xi} \geq 0$, $v_n \geq 0$.

3.2.1 A potential higher order scheme

Lemma 1 of [Mac11] gives the following construction of a positive discrete random variable whose moments match the first four moments of X.

Lemma 3.1 (Mackevičius)**.** *Let X be a nonnegative random variable nonconcentrated in $\{0,1\}$ with $m_i := \mathbb{E}\left[X^i\right] < \infty$, $i = 1,2,3,4$.*

1. *Let*
$$s := \frac{m_1 m_4 - m_2 m_3}{m_1 m_3 - m_2^2}; \qquad q := \frac{m_2 m_4 - m_3^2}{m_1 m_3 - m_2^2};$$
Then $s > 0$, $q > 0$, $d := s^2 - 4q > 0$, and the polynomial $P(x) = x^2 - sx + q$ has two positive roots
$$x_{\pm} = \frac{1}{2} s \pm \sqrt{d}.$$

2. $$p_- := \frac{m_1 x_+ - m_2}{x_-(x_+ - x_-)} > 0; \quad p_+ := \frac{m_2 - m_1 x_-}{x_+(x_+ - x_-)} > 0; \quad p_0 = 1 - p_+ - p_- > 0.$$

3. *Let $\hat{X}$ be a random variable taking the values $x_+, x_-, 0$ with probabilities p_+, p_-, p_0 respectively. Then*
$$\mathbb{E}\left[\hat{X}^i\right] = m_i, \quad i = 1,2,3,4.$$

4 Weak error analysis

Consider first $\mathbb{E}\left[g(v_\Delta)\right]$ where g is suitably smooth. To compute the weak error of the discretization $\hat{v}$, we Taylor expand around $\hat{\xi}_\Delta$ to get

$$\mathbb{E}\left[g(\hat{v}_\Delta) - g(\hat{\xi}_\Delta)\right] = \sum_{k=1}^{\nu} \frac{g^{(k)}(\hat{\xi}_\Delta)}{k!} \mathbb{E}\left[(\hat{v}_\Delta - \hat{\xi}_k)^k)\right] + \text{ higher order }.$$

Similarly

$$\mathbb{E}\left[g(v_\Delta) - g(\hat{\xi}_\Delta)\right] = \sum_{k=1}^{\nu} \frac{g^{(k)}(\hat{\xi}_\Delta)}{k!} \mathbb{E}\left[(v_\Delta - \hat{\xi}_k)^k\right] + \text{ higher order }.$$

In a rough affine forward variance model $\mathbb{E}_t\left[(v_\Delta - \hat{\xi}_k)^k)\right] \sim \Delta^{kH}$. Moreover, Remark A.2 makes us believe that we cannot compute such moments with an error of less than $\mathcal{O}(\Delta^{1+2H})$. It follows, for the kind of scheme we have in mind, $\nu H \leq 1 + 2H$ so that

$$\nu < \frac{1+2H}{H}.$$

Assuming $H \leq 1/4$, this gives $\nu \leq 6$. So it is certainly justified to try the higher order scheme of Mackevičius.

5 An Euler-like scheme for v

We simulate the $u_k := v_k - \hat{\xi}_k$ Euler-style as

$$u_k = \int_{(k-1)\Delta}^{k\Delta} g(k\Delta - s)\ \sqrt{v_s}\, dW_s \approx b_1^\star \chi_k$$

where as before,

$$\chi_k = \int_{(k-1)\Delta}^{k\Delta} \sqrt{v_s}\, dW_s$$

and the $b^\star$ are chosen to be the root mean square average of g over the interval:

$$b_{n-k+1}^\star = \sqrt{\frac{1}{\Delta}\int_{(k-1)\Delta}^{k\Delta} g^2(n\Delta - s)\, ds}. \tag{5.1}$$

$b^\star$ may be re-expressed more conveniently in the form

$$b_j^{\star 2} = \frac{1}{\Delta}\left[\mathcal{G}_{0,0}(j\Delta) - \mathcal{G}_{0,0}((j-1)\Delta)\right].$$

Our Euler-like scheme then reads

$$v_n^E = \xi_n + \sum_{k=1}^{n} b_{n-k+1}^\star \chi_k. \tag{5.2}$$

With this definition of $b^\star$ [4], we have that

$$\mathrm{var}[v_T^E] = \sum_{k=1}^{n} {b_{n-k+1}^\star}^2\, \mathrm{var}(\chi_k) \to \int_0^T \xi(s)\, g^2(n\Delta - s)\, ds = \mathrm{var}[v_T],$$

as $\Delta = T/n \to 0$.

[4] This differs from the choice made in [BLP17].

5.1 Computation of $\hat{\xi}_n = \mathbb{E}\left[\, v_n | \mathcal{F}_{n-1}\right]$

It is immediate that (5.2) may be rewritten as $v_n^E = \hat{\xi}_n + u_n$ with

$$\hat{\xi}_n = \mathbb{E}\left[v_n|\mathcal{F}_{n-1}\right] = \mathbb{E}\left[\xi_n + \sum_{k=1}^{n-1} b^\star_{n-k+1}\chi_k \middle| \mathcal{F}_{n-1}\right]. \tag{5.3}$$

Define $u_n = v_n^E - \hat{\xi}_n$. Then $\mathrm{var}[u_n] = b_1^{\star 2}\,\mathrm{var}[\chi_n]$. The essence of the Euler-discretization of [BLP17] is that the χ_n and the u_n are perfectly correlated. Moreover, both χ_n and u_n are zero-mean random variables so we can write

$$\chi_n = \frac{u_n}{b_1^\star}.$$

Then, in summary, starting with $\hat{\xi}_1 = \xi_1$, the simulation scheme is

$$\begin{aligned} v_n^E &= \hat{\xi}_n + u_n \\ \hat{\xi}_n &= \xi_n + \sum_{k=1}^{n-1} \frac{b^\star_{n-k+1}}{b_1^\star}\, u_k. \end{aligned} \tag{5.4}$$

Finally, note that since $\mathrm{var}[u_k] = \bar{v}_{k-1}\,\mathcal{G}_{0,0}(\Delta)$,

$$\begin{aligned} \mathrm{var}[v_n^E] &= \sum_{k=1}^{n} \mathbb{E}\left[\bar{v}_{k-1}\right]\, \left[\mathcal{G}_{0,0}((n-k+1)\Delta) - \mathcal{G}_{0,0}((n-k)\Delta)\right] \\ &\approx \sum_{k=1}^{n} \mathbb{E}\left[\bar{v}_{k-1}\right]\, g((n-k)\Delta)\,\Delta \\ &\to \int_0^T g(T-s)\,\xi(s)\,ds \text{ as } n \to \infty. \end{aligned}$$

Remark 5.1. *The covariance structure is obviously not correctly matched though.*

5.2 Positive simulation schemes

Obviously, according to the scheme (5.2), depending on how we sample the random variables χ_n, we could have $v_n^E < 0$, which is obviously undesirable and inconsistent with the process being simulated. We thus choose to simulate v_n^E using Andersen's QE scheme as explained in [Alf15] or the positive discrete scheme of Mackevičius. In each of these cases, at each time step, we need the expectation and variance of the v process, conditional on $\mathcal{F}_{(n-1)\Delta}$.

$\hat{\xi}_n = \mathbb{E}\left[\, v_n | \mathcal{F}_{n-1}\right]$ is given by (5.4) and from (A.2), we have the approximation

$$\mathrm{var}[v_n|\mathcal{F}_{n-1}] \approx \frac{\mathcal{G}_{0,0}(\Delta)}{2H+1}\,\left(\hat{\xi}_n + 2H\, v_{n-1}\right) =: \bar{v}_{n-1}\,\mathcal{G}_{0,0}(\Delta). \tag{5.5}$$

Example 5.1 (Gamma kernel)**.** *Let*

$$g(\tau) = \sqrt{2H}\,\eta\,\tau^{\alpha-1}\,e^{-\lambda\tau} =: \tilde{\eta}\,\tau^{\alpha-1}\,e^{-\lambda\tau}, \tag{5.6}$$

which has power-law and exponential kernels as special cases. Then

$$\begin{aligned}\Delta\, b_j^{\star 2} &= \mathcal{G}_{0,0}(j\Delta) - \mathcal{G}_{0,0}((j-1)\Delta) \\ &= \frac{\tilde{\eta}^2}{(2\lambda)^{2H}}\,\{\Gamma(2H, 2\lambda(j-1)\Delta) - \Gamma(2H, 2\lambda j\Delta)\}\,.\end{aligned}$$

In the special case $\lambda = 0$, this simplifies to $\Delta\, b_j^{\star 2} = \eta^2\,\Delta^{2H}$.

5.3 Simulating x_T

Let $\bar{v}_n = \frac{1}{2}(v_n + v_{n-1})$. Then, given v, we simulate x as

$$\begin{aligned}x_n &= x_{n-1} - \tfrac{1}{2}\,\bar{v}_n\,\Delta + \sqrt{1-\rho^2}\,\sqrt{\bar{v}_n\,\Delta}\,Z_n^{\perp} + \rho\,\chi_n \\ &\approx x_{n-1} - \tfrac{1}{2}\,\bar{v}_n\,\Delta + \sqrt{1-\rho^2}\,\sqrt{\bar{v}_n\,\Delta}\,Z_n^{\perp} + \rho\,\frac{u_n}{b_1^{\star}}.\end{aligned} \tag{5.7}$$

where $Z^{\perp} \sim N(0,1)$ and $u_n = v_n - \hat{\xi}_n$.

Remark 5.2. *Note that, assuming that $\bar{v}_n\Delta - \int_{t_{n-1}}^{t_n} v_s\,ds$ tends to zero fast enough, the martingale correction of [And07] is higher order.*

5.4 Check of $Y \diamond M$

One constraint is that in order to match the volatility skew, we would like to be able to match the covariance of $Y_T = \int_0^T \sqrt{v_s}\,dW_s$ and $\mathcal{V}_T := \int_0^T v_s\,ds$. Recall that $\mathrm{cov}[Y_T, \mathcal{V}_T] = (Y \diamond M)_0(T)$ and that

$$(Y \diamond M)_0(T) = \int_0^T \xi(s)\,ds \int_s^T g(u-s)\,du.$$

Our discretization (5.7) gives

$$Y_N = \sum_{j=1}^{N} (Y_j - Y_{j-1}) = \sum_{j=1}^{N} \frac{u_j}{b_1^{\star}}.$$

Also, recall that

$$v_n^E = \xi_n + \sum_{k=1}^{n} \frac{b_{n-k+1}^{\star}}{b_1^{\star}}\,u_k.$$

so that

$$\mathcal{V}_N = \sum_{n=1}^{N} \xi_n \, \Delta + \sum_{n=1}^{N} \Delta \sum_{k=1}^{n} \frac{b^\star_{n-k+1}}{b^\star_1} \, u_k.$$

Thus, with $T = N\Delta$ and using that $\mathrm{cov}[u_j, u_k \approx \xi_k \, \delta_{jk} \, b_1^{\star 2} \, \Delta$ and that $b^\star_j \approx g(j\Delta)$,

$$\begin{aligned}
\mathrm{cov}[Y_T, \mathcal{V}_T] &= \sum_{n=1}^{N} \Delta \sum_{k=1}^{n} \xi_k \, b^\star_{n-k+1} \, \Delta \\
&\approx \sum_{n=1}^{N} \Delta \int_0^{n\Delta} \xi(s) \, g(n\Delta - s) \, ds \\
&\approx \int_0^T du \int_0^u \xi(s) \, g(u-s) \, ds \\
&\to \int_0^T \xi(s) \, ds \int_s^T g(u-s) \, du \text{ as } \Delta \to 0.
\end{aligned}$$

We conclude that (5.7) is a good consistent discretization.

5.5 Numerical results

In Figure 5.1 we compare 1-year smiles generated with the above QE scheme with those generated using the Adams scheme with 500 steps. Parameters used were:

$$\xi(u) = 0.025; \; H = 0.05; \; \nu = 0.4; \rho = -0.65.$$

Note that ν in the original formulation of the rough Heston model and η in (5.6) are related as

$$\eta = \frac{\nu}{\sqrt{2H} \, \Gamma(\alpha)}.$$

6 A QE hybrid scheme

The scheme (5.2) matches means and variances at each step but it does not match the covariance structure of the process. For example, consider the covariance between v_1 and $\hat{\xi}_2$ which from (2.2) is given approximately by

$$\mathrm{cov}[v_1, \hat{\xi}_2] = \bar{\xi} \, \mathcal{G}_{0,1}(\Delta).$$

(5.2) and (5.4) approximate this covariance as $\mathrm{cov}[v_1, \hat{\xi}_2] \approx \bar{\xi} \, b_2^\star \, b_1^\star \, \Delta$ which is equivalent to the approximation

$$\left(\int_0^\Delta g(s) \, g(s+\Delta) \, ds \right)^2 \approx \left(\int_0^\Delta g^2(s) \, ds \right) \left(\int_0^\Delta g^2(s+\Delta) \, ds \right), \tag{6.1}$$

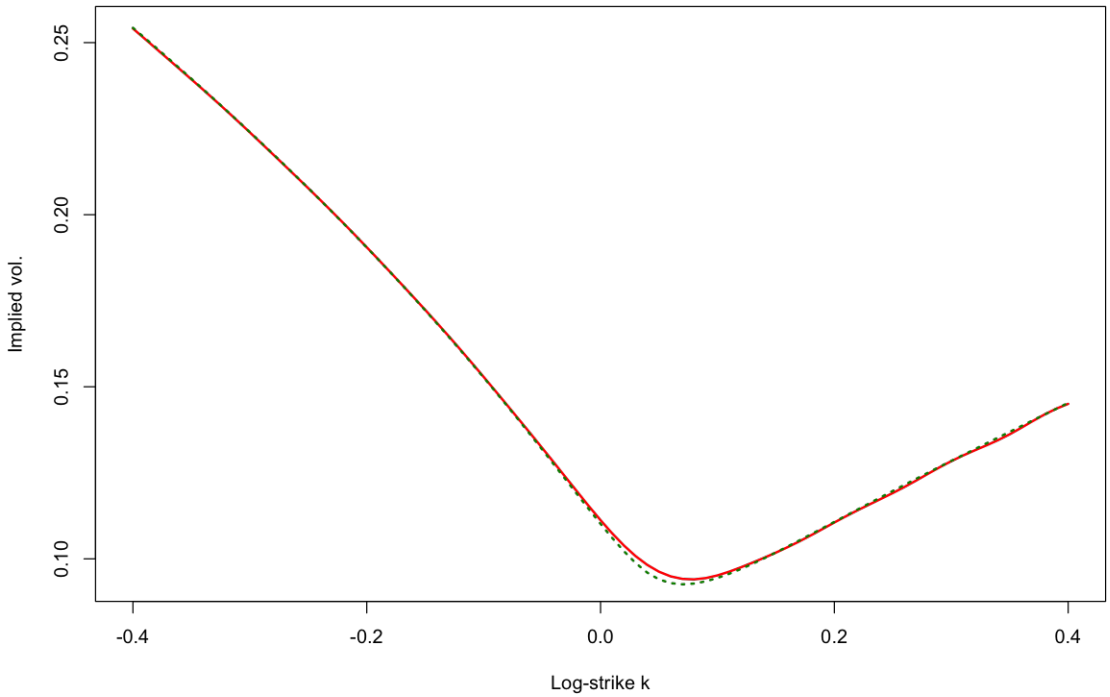

Figure 5.1: The red line is from the Adams scheme with 5,000 steps. The blue dashed line is from the QE simulation using 100,000 paths and 1,000 steps. We see numerical evidence of convergence!

which, though exact in the case of an exponential kernel, is obviously inaccurate for small H. The essence of the hybrid scheme with $\kappa = 1$ of [BLP17] is to correct the error in the approximation (6.1) by simulating another random variable that is uncorrelated with changes in v_k so as to match the covariance of v_1 and v_2.

One could add further corrections so as to correct (for example) the covariance of v_1 and v_3. However, for small H such higher order corrections are small and seem not to improve the performance of the algorithm[5]. This is as we would expect because the decay of the autocorrelation function is very fast for small H; we would only see the impact of higher order corrections if we were to choose very tiny time steps.

6.1 A 2-D version of the QE step

Let

$$\begin{aligned} u_1 &= \int_0^\Delta g(\Delta - s)\, \sqrt{v_s}\, dW_s \\ \chi_1 &= \int_0^\Delta \sqrt{v_s}\, dW_s. \end{aligned}$$

Linear regression gives

$$u_1 = \beta_{v\chi}\, \chi_1 + \epsilon_1$$

where ϵ and χ are independent. From (2.2),

$$\beta_{v\chi} = \frac{\mathcal{G}_{0,0}(\Delta)}{\Delta} = b_1^\star.$$

Again from (2.2), we have

$$\begin{aligned} \mathrm{var}[u_1] &= \bar{\xi}\, \mathcal{G}_{0,0}(\Delta) \\ \mathrm{var}[\chi_1] &= \bar{\xi}\, \Delta \\ \mathrm{var}[\epsilon_1] &= \mathrm{var}[u_1] - \beta_{v\chi}^2\, \mathrm{var}[\chi_1] = \bar{\xi}\, \mathcal{G}_{0,0}(\Delta)\, (1 - \rho_{v\chi}^2). \end{aligned}$$

Let's generate both χ_1 and ϵ_1 separately using the QE scheme. In particular, since $v_1 \geq 0$, we must have

$$\beta \chi_1 + \epsilon_1 + \xi_1 \geq 0.$$

Thus for some $a > 0$, we must have

$$\beta \chi_1 + a\, \xi_1 \geq 0; \quad \epsilon_1 + (1 - a)\, \xi_1 \geq 0.$$

That gives

$$\begin{aligned} \mathbb{E}\left[\beta \chi_1 + a\, \xi_1\right]^2 &= a^2\, \xi_1^2 = \psi_\chi\, \beta^2\, \bar{\xi}\, \Delta = \psi_\chi\, \rho_{v\chi}^2\, \bar{\xi}\, \mathcal{G}_{0.0}(\Delta) \\ \mathbb{E}\left[\epsilon_1 + (1 - a)\, \xi_1\right]^2 &= (1 - a)^2\, \xi_1^2 = \psi_\epsilon\, \bar{\xi}\, \mathcal{G}_{0,0}(\Delta)\, (1 - \rho_{v\chi}^2). \end{aligned}$$

[5] I conjecture that we can prove that higher order corrections are negligible for H small.

Once again approximating $\bar{\xi} = \xi_1$, and solving for a, we find that there is a solution only when $a = \frac{1}{2}$. Thus we need to solve the system

$$\begin{aligned} \tfrac{1}{4}\xi_1^2 &= \psi_\chi \bar{\xi}\, \mathcal{G}_{0,0}(\Delta)\, \rho_{v\chi}^2 \\ \tfrac{1}{4}\xi_1^2 &= \psi_\epsilon \bar{\xi}\, (1-\rho_{v\chi}^2)\, \mathcal{G}_{0,0}(\Delta). \end{aligned}$$

This gives

$$\psi_\chi = \frac{\bar{\xi}_1}{4\,\mathcal{G}_{0,0}(\Delta)\,\rho_{v\chi}^2}; \quad \psi_\epsilon = \frac{\bar{\xi}_1}{4\,\mathcal{G}_{0,0}(\Delta)\,(1-\rho_{v\chi}^2)}. \tag{6.2}$$

We have now, at least in principle, generated the non-negative random variables χ_1 and ϵ_1 and so we also have $v_1 = \xi_1 + \beta_{v\chi}\, u_1 + \epsilon_1$.

6.1.1 A first small H approximation

According to the approximate covariance matrix (2.5), $\hat{\xi}_2$ is perfectly correlated with χ_1. Moreover $\mathbb{E}\left[\chi_1\right] = 0$ so we may write

$$\hat{\xi}_2 - \xi_2 \approx \frac{\sqrt{\mathrm{var}[\hat{\xi}_2]}}{\sqrt{\mathrm{var}[\chi_1]}}\chi_1 \approx \sqrt{\frac{\mathcal{G}_{1,1}(\Delta)}{\Delta}}\chi_1 = b_2^\star\, \chi_1. \tag{6.3}$$

Remark 6.1. *We can come back and generalize this step later. But we expect the correction to be very small.*

6.2 The generic step

Recall that in the Euler-style case,

$$v_n^E = \xi_n + \sum_{k=1}^{n} b_{n-k+1}^\star\, \chi_k.$$

In analogy with [BLP17], we write our proposed hybrid scheme as

$$v_n^H = \xi_n + u_n + \sum_{k=1}^{n-1} b_{n-k+1}^\star\, \chi_k,$$

where u_n is no longer approximated as $b_1^\star\, \chi_n$. Thus for $n > 1$, given $\mathcal{F}_{n-1}$, dropping the superscript H for ease of notation, generate χ_n and ϵ_n together using the 2-D QE scheme with

$$\begin{aligned} \mathbb{E}\left[\beta\chi_n|\mathcal{F}_{n-1}\right] &= \tfrac{1}{2}\hat{\xi}_n; & \mathbb{E}\left[\epsilon_n|\mathcal{F}_{n-1}\right] &= \tfrac{1}{2}\hat{\xi}_n; \\ \mathrm{var}[\chi_n|\mathcal{F}_{n-1}] &= \bar{\xi}_n\,\Delta; & \mathrm{var}[\epsilon_n|\mathcal{F}_{n-1}] &= \bar{\xi}\,\mathcal{G}_{0,0}(\Delta), \end{aligned}$$

and

$$\bar{\xi}_n = \frac{1}{2H+1}\left[\hat{\xi}_n + 2H\, v_{n-1}\right].$$

Then with[6]

$$\beta = \frac{\mathcal{G}_{0,1}(\Delta)}{\mathcal{G}_{0,0}(\Delta)}; \quad \rho = \frac{\mathcal{G}_{0,1}(\Delta)}{\sqrt{\mathcal{G}_{0,0}(\Delta)}\sqrt{\mathcal{G}_{1,1}(\Delta)}}; \quad b_j^\star = \sqrt{\frac{\mathcal{G}_{0,0}(j\Delta) - \mathcal{G}_{0,0}((j-1)\Delta)}{\Delta}},$$

compute $\hat{\xi}_{n+1}$ as

$$\hat{\xi}_{n+1} = \xi_{n+1} + \sum_{k=1}^{n} b_{n-k+1}^\star \chi_k. \tag{6.4}$$

Remark 6.2. *After implementing* (6.4)*, it seems that* $\hat{\xi}_{n+1} > 0$*. Maybe we can prove this?*

6.3 Numerical results

In Figure 6.1 we compare 1-year smiles generated with the above hybrid QE scheme with those generated using the Euler-like scheme of Section 5 and using the Adams scheme with 5,000 steps. Again, the parameters used were:

$$\xi(u) = 0.025;\ H = 0.05;\ \nu = 0.4; \rho = -0.65.$$

Note that ν in the original formulation of the rough Heston model and η in (5.6) are related as

$$\eta = \frac{\nu}{\sqrt{2H}\,\Gamma(\alpha)}.$$

In Figure 6.2, we see that 200 steps using the hybrid scheme seems to be a good compromise between speed and accuracy. Note that a back-of-the-envelope estimate says that a 200 step computation should run approximately 25 times faster than a 1,000 step one.

6.4 Probabilistic interpretation of the hybrid scheme

The Euler scheme can be seen as simulation of the rough process as if it were a semimartingale. The hybrid scheme (with $\kappa = 1$) adds independent noise at each step. In some sense the hybrid scheme represents the rough process as a semimartingale plus independent noise. Indeed, that a semimartingale sampled with (independent) measurement error may look rough has been noticed before, see the detailed discussion in [FTW19].

In the limit of many components, the Markovian approximation of [**?**] corresponds to the Euler scheme (5.2). As pointed out in that paper, the simulated process is a semimartingale that scales as a rough process as the sampling interval increases. Adding the independent noise $Z^\perp$ as each step restores apparent roughness for every sampling interval.

[6] Look at $b^\star$ as the root mean square average of g to see the expression for $b^\star$.

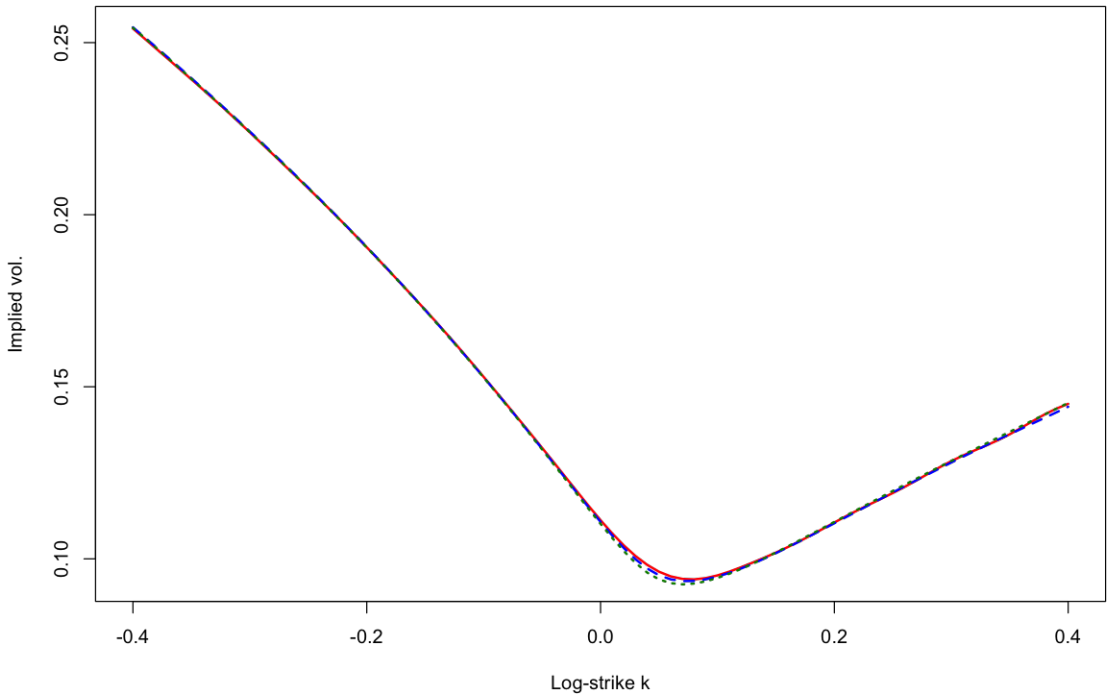

Figure 6.1: The red line is from the Adams scheme with 5,000 steps. The blue dashed line is from the hybrid QE simulation using 100,000 paths and 1,000 steps; the green dotted line is from the Euler QE simulation with 100,000 paths and 1,000 steps. We see that the hybrid scheme is marginally but significantly better.

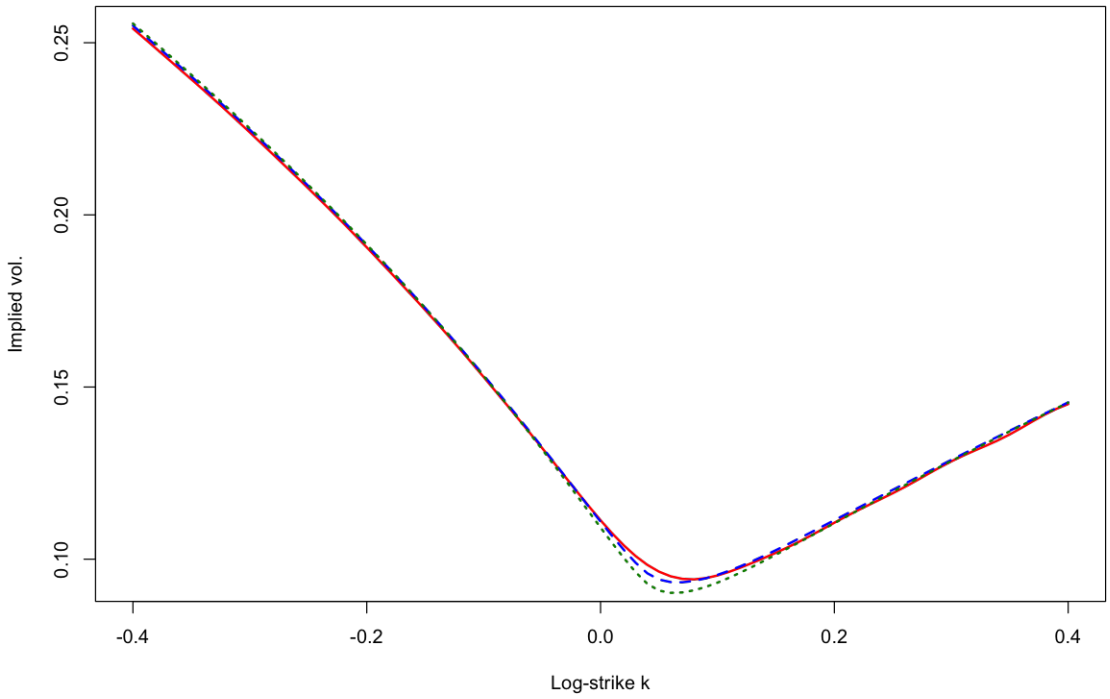

Figure 6.2: The red line is from the Adams scheme with 5,000 steps. The blue dashed line is from the hybrid QE simulation using 100,000 paths and 200 steps; the green dotted line is from the Euler QE simulation with 100,000 paths and 200 steps. We see that the hybrid scheme is marginally but significantly better.

6.5 Financial interpretation of the hybrid scheme

Since $v_0 = \xi_0(0)$ is given, the first step is almost exact.

$$v_1 = \xi_1 + \eta \sqrt{w(\Delta)} \sqrt{v_0} Z_1.$$

The next step is

$$\begin{aligned} v_2 &= \xi_2 + \int_0^{2\Delta} g(2\Delta - s) \sqrt{v_s} \, dW_s \\ &= \xi(2\Delta) + \int_0^{\Delta} g(2\Delta - s) \sqrt{v_s} \, dW_s + \int_{\Delta}^{2\Delta} g(2\Delta - s) \sqrt{v_s} \, dW_s \\ &\approx \left\{ \xi_2 + \sqrt{v_0} \int_0^{\Delta} g(2\Delta - s) \, dW_s \right\} + \sqrt{v_1} \int_0^{\Delta} g(s) \, dW_s. \end{aligned}$$

The term in brackets, represent the forward variance curve after one simulation step. That is

$$\mathbb{E}_{\Delta}[v_2] = \xi_2 + \sqrt{v_0} \int_0^{\Delta} g(2\Delta - s) \, dW_s.$$

The hybrid correction term v^{corr} in (**??**) corresponds to computing the expectation of

$$\int_0^{\Delta} g(2\Delta - s) \sqrt{v_s} \, dW_s$$

conditional on $\int_0^{\Delta} g(\Delta - s) \, dW_s$ by linear regression. Generalizing this argument, we may rewrite the hybrid scheme (**??**) as

$$v_n^H = \hat{\xi}_n + \sqrt{v_{n-1}} \, b_1^{\star} \sqrt{\Delta} \left(\rho_1 Z_n + \bar{\rho}_1 Z_n^{\perp} \right) \tag{6.5}$$

with

$$\hat{\xi}_n := \mathbb{E}_{(n-1)\Delta}[v_n] = \xi_n + \sum_{k=1}^{n-1} \sqrt{v_{k-1}} \, b_{n-k+1}^{\star} \sqrt{\Delta} Z_k.$$

In words, at time $(n-1)\Delta$, add a martingale increment to the expectation of v_n given $\mathcal{F}_{(n-1)\Delta}$. We see explicitly that $\mathbb{E}_{(n-1)\Delta}[v_n]$ takes into account the whole history of the increments $Z_k, k < n$.

7 Numerical tests of convergence

After extensive testing, it appears that the order of weak convergence of the hybrid scheme is one. That being the case, we may assume that the exact smile is given by Richardson extrapolation of the smiles computed with 1,000 and 500 time steps respectively. This allows estimation of the bias of the hybrid scheme as a function of the number of time steps. In Figure 7.1, we plot smiles with various numbers of time steps and plot errors vs time steps. Order one weak convergence is clearly demonstrated.

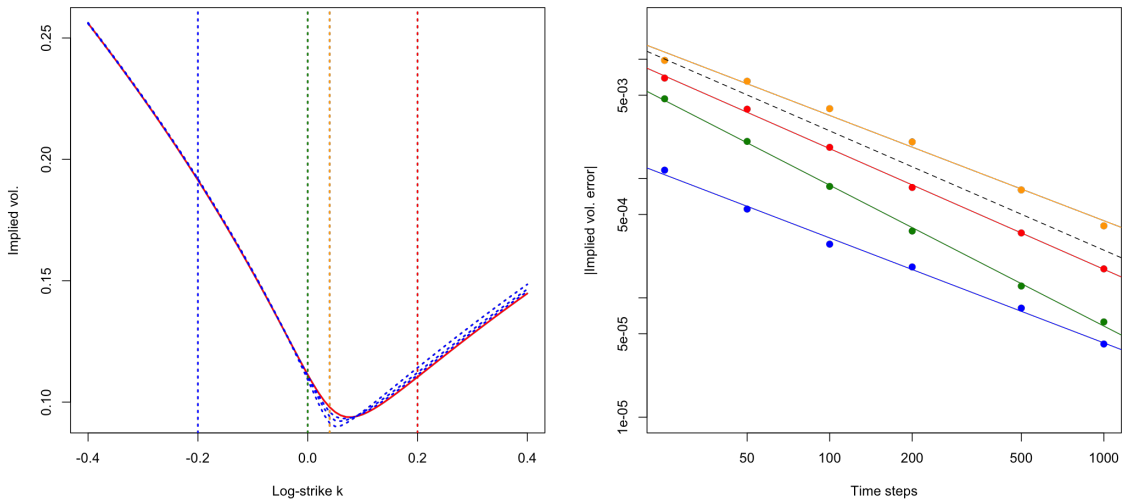

Figure 7.1: In the LH plot, the red line is from Richardson extrapolation of 1,000 and 500 time steps smiles. The blue dotted lines are smiles computed with various lower numbers of time steps. In the RH plot, we plot absolute implied volatility errors vs time steps, for various log-strikes: $k \in \{-0.2, 0, 04, 0.2\}$ with colors corresponding to the vertical lines in the LH plot. The dashed black line with slope -1 is plotted for reference, clearly demonstrating order one weak convergence. All simulations are with 10 million time steps.

7.1 Richardson extrapolation

Given the apparent order one weak convergence of the hybrid scheme, we may compute various Richardson extrapolated smiles and plot their estimated errors. We plot results in Figure 7.2.

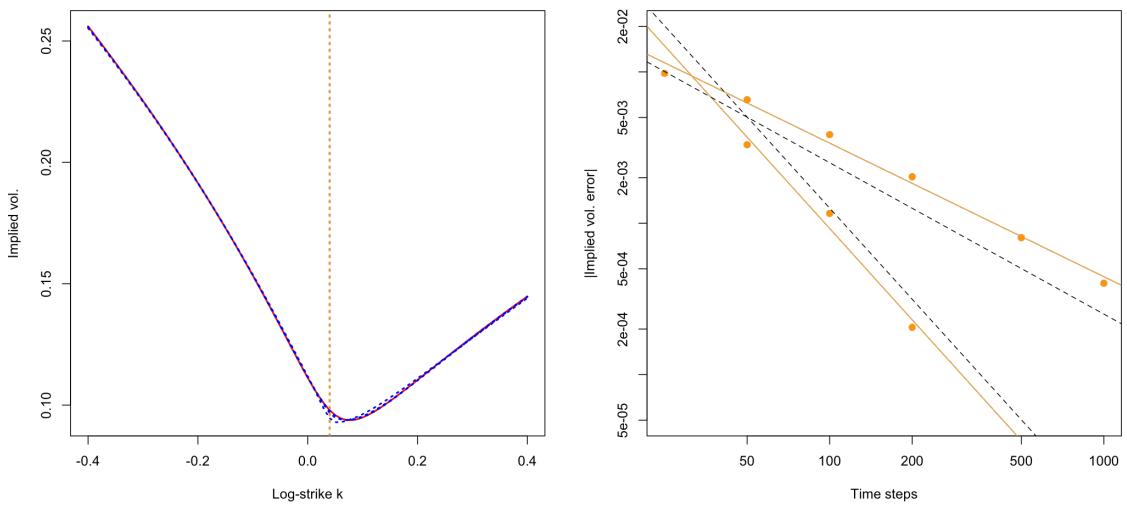

Figure 7.2: In the LH plot, the red line is from Richardson extrapolation of 1,000 and 500 time steps smiles. The blue dotted lines are smiles computed using Richardson extrapolation with various lower numbers of time steps. In the RH plot, we plot absolute implied volatility errors vs time steps, for log-strike $k = 0.04$, the orange vertical line in the LH plot, where errors are maximized. Errors without extrapolation are superimposed for reference as are black-dashed lines with slopes -1 and -2 respectively. We see evidence of order 2 weak convergence of Richardson-extrapolated smiles. All simulations are with 10 million time steps.

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A Approximate variance

For a kernel g that is the product of a power-law and a regularly varying function, we have the following very natural-looking approximation of the variance over a timestep.

Lemma A.1. *Let*

$$\mathcal{G}_{0,0}(\Delta) = \int_0^\Delta g(s)^2\, ds.$$

Then

$$\mathrm{var}[v_\Delta] \approx \frac{\mathcal{G}_{0,0}(\Delta)}{2H+1}\, \left[\xi(\Delta) + 2H\, v_0\right]. \tag{A.1}$$

Proof. We approximate the variance after one step as follows.

$$\begin{aligned}
\mathrm{var}\,[v_\Delta] &= \int_0^\Delta \xi(s)\, g(\Delta - s)^2\, ds \\
&= \xi(\Delta) \int_0^\Delta g(\Delta - s)^2\, ds - \int_0^\Delta \xi'(s)\, ds \int_0^s g(\Delta - r)^2\, dr \\
&\approx \xi(\Delta) \int_0^\Delta g(s)^2\, ds - \left[\xi(\Delta) - v_0\right] \frac{1}{\Delta} \int_0^\Delta ds \int_0^s g(\Delta - r)^2\, dr \\
&= \xi(\Delta) \int_0^\Delta g(s)^2\, ds - \left[\xi(\Delta) - v_0\right] \frac{1}{\Delta} \int_0^\Delta g(s)^2\, s\, ds.
\end{aligned}$$

Now let $g(\tau) = \tau^{H-1/2}\, L_g(\tau)$, $\tau \in (0, T]$. Then, for some $s^\star \in [0, T]$,

$$\mathrm{var}\,[v_\Delta] \approx \xi(\Delta)\, L_g(s^\star)^2\, \frac{\Delta^{2H}}{2H} - \left[\xi(\Delta) - v_0\right]\, L_g(s^\star)^2\, \frac{\Delta^{2H}}{2H+1}.$$

Also

$$L_g(s^\star)^2 \, \frac{\Delta^{2H}}{2H} \approx \int_0^\Delta g(s)^2 \, ds = \mathcal{G}_{0,0}(\Delta),$$

which completes the proof. □

Remark A.1. *Note that in the limit $H \to 0$, all of the weight is on the end of the interval.*

Remark A.2. *From the above proof, we see that given sufficient smoothness of ξ,*

$$\mathrm{var}\,[v_\Delta] = \frac{\mathcal{G}_{0,0}(\Delta)}{2H+1} \, \left[\xi(\Delta) + 2H \, v_0\right] + O(\Delta^{2H+1}).$$

Similarly, we have the following lemma.

Lemma A.2. *Let*

$$v_\Delta = \xi + \int_0^\Delta g(\Delta - s) \, \sqrt{v_s} \, dW_s, \quad \chi_\Delta = \int_0^\Delta \sqrt{v_s} \, dW_s.$$

Further let

$$\mathcal{G}_0(\Delta) = \int_0^\Delta g(s) \, ds.$$

Then

$$\mathrm{cov}[v_\Delta, \chi_\Delta] \approx \frac{\mathcal{G}_0(\Delta)}{\alpha+1} \, \left[\xi(\Delta) + \alpha \, v_0\right]. \tag{A.2}$$

Proof. We approximate this covariance after one step as follows.

$$\begin{aligned}
\mathrm{cov}[v_\Delta, Y_\Delta] &\approx \int_0^\Delta \xi(s) \, g(\Delta - s) \, ds \\
&= \xi(\Delta) \int_0^\Delta g(\Delta - s) \, ds - \int_0^\Delta \xi'(s) \, ds \int_0^s g(\Delta - r) \, dr \\
&\approx \xi(\Delta) \int_0^\Delta g(s) \, ds - \left[\xi(\Delta) - v_0\right] \, \frac{1}{\Delta} \int_0^\Delta ds \int_0^s g(\Delta - r) \, dr \\
&= \xi(\Delta) \int_0^\Delta g(s) \, ds - \left[\xi(\Delta) - v_0\right] \, \frac{1}{\Delta} \int_0^\Delta g(s) \, s \, ds.
\end{aligned}$$

Now let $g(\tau) = \tau^{H-1/2} \, L_g(\tau)$, $\tau \in (0, T]$. Then, for some $s^\star \in [0, T]$,

$$\mathrm{cov}[v_\Delta, Y_\Delta] \approx \xi(\Delta) \, L_g(s^\star) \, \frac{\Delta^\alpha}{\alpha} - \left[\xi(\Delta) - v_0\right] \, L_g(s^\star) \, \frac{\Delta^\alpha}{\alpha+1}.$$

Also

$$L_g(s^\star) \, \frac{\Delta^\alpha}{\alpha} \approx \int_0^\Delta g(s) \, ds = \mathcal{G}_0(\Delta),$$

which completes the proof. □

B Bell polynomials

$$B_{n,k}(x_1, x_2, \ldots, x_{n-k+1}) = \sum \frac{n!}{j_1! j_2! \cdots j_{n-k+1}!} \left(\frac{x_1}{1!}\right)^{j_1} \left(\frac{x_2}{2!}\right)^{j_2} \cdots \left(\frac{x_{n-k+1}}{(n-k+1)!}\right)^{j_{n-k+1}},$$

where the sum is taken over all sequences $j_1, j_2, j_3, ..., J_{n-k+1}$ of non-negative integers such that these two conditions are satisfied:

$$\begin{aligned} j_1 + j_2 + \cdots + j_{n-k+1} &= k, \\ j_1 + 2j_2 + 3j_3 + \cdots + (n-k+1)j_{n-k+1} &= n. \end{aligned}$$

The sum

$$B_n(x_1, \ldots, x_n) = \sum_{k=1}^{n} B_{n,k}(x_1, x_2, \ldots, x_{n-k+1})$$

is called the nth "complete exponential Bell polynomial".